Work-let Name: On device Topic modelling for small text data

Worklet Details

1. Worklet ID: 25PI05TU
2. College Name: Thapar Institute of Engineering & Technology

KPIs achieved till now

- Successfully fine-tuned the MobileBERT baseline architecture by framing SMS keyword extraction as a Token Classification task.
- MobileBERT achieved strong baseline performance with Accuracy 89.47%, Precision 84.24%, and Recall 92.55%.
- Implemented and fine-tuned a larger DistilBERT-based model for improved keyword extraction accuracy.
- DistilBERT achieved significantly better validation performance with Accuracy 97.44%, Keyword F1-Score 97.01%, Precision 96.23%, and Recall 97.81%.

Any Challenges/ Issues faced

- **No Ground-Truth Token Labels:** The dataset does not provide token-level keyword annotations, which makes supervised training difficult. We addressed this using RAKE-based pseudo-labelling to create distant supervision signals and then trained the Transformer model to generalise beyond those heuristic labels.
- **Class Imbalance in Keyword Extraction:** Non-keyword tokens typically dominate keyword tokens, which can bias optimization. We mitigated this using inverse-frequency class weighting during training so the model pays sufficient attention to the keyword class.

Next Steps

- **Conclude Current Worklet**: Finalize and close out the active model evaluation worklet.
- **Optimize and Fine-Tune Model**: Tune the model parameters and architecture to align precisely with specific project requirements and performance needs.

Profile	Accuracy	Keyword F1	Keyword Recall	Macro F1
mobilebert_baseline	0.8947	0.8820	0.9255	0.8935
distilbert_accuracy	0.9744	0.9701	0.9781	0.9739

Work-let Name: On device Topic modelling for small text data

Worklet Details

1. Worklet ID: Worklet ID: 25PI05TU
2. College Name: Thapar Institute of Engineering & Technology

Resource Requirement Identified with Status

- Compute Resources for Training: GPU environment for fine-tuning MobileBERT (Status: Secured/Completed).
- Testing Hardware: Android device for testing on-device inference latency and real-world memory profiling (Status: Identified / In Progress).

Github Repository Link and PRISM Portal Registration Status

Github Repository Link - https://github.com/priyanshigoyal2801/25PI05TU
Portal Registration Status - Completed

Any Anticipated Break (Due to exams etc)

Internship Period - June & July

Data Collection Information

- **Source Type:** Open Source
- **Dataset Name:** UCI SMS Spam Collection Dataset (~5,574 English SMS messages).
- **Details**: Data is automatically downloaded and parsed by our download_data.py script. We apply RAKE algorithm for unsupervised pseudo-labeling.
- Dataset Link: https://archive.ics.uci.edu/ml/datasets/sms+spam+collection

Date: 26-06-2026

Work-let Name: On device Topic modelling for small text data

Worklet Details

1. Worklet ID: 25PI05TU
2. College Name: Thapar Institute of Engineering & Technology

Professor Comments (To be filled in and presented by professor (Either)

Status and future plan
Latest Literature Review
Expert Insights
Any other comment

COMPUTATION RESOURCE Details:

Local PC (Laptop with dedicated GPU). Total compute hours: ~0.5 to 1 hour for full end-to-end dataset preprocessing, model fine-tuning (5 epochs), and TFLite conversion.

Date: 26-06-2026

Sample Outputs:

```
================================================================================
Device      : CPU
Model       : Fine-tuned DistilBERT
Task        : SMS Keyword Extraction
Dataset     : UCI SMS Spam Collection + RAKE weak supervision
Best Score  : Accuracy 97.44% | Keyword F1 97.01%
================================================================================

Example 1
--------------------------------------------------------------------------------
Input SMS:
Your OTP is 482910. Valid for 10 minutes.

Extracted Keywords:
  - OTP                         confidence=0.98
  - 482910                      confidence=0.97
  - 10 minutes                  confidence=0.94

Example 2
--------------------------------------------------------------------------------
Input SMS:
Your HDFC account was debited Rs 5000 at Amazon Pay.

Extracted Keywords:
  - HDFC account                confidence=0.96
  - debited                     confidence=0.95
  - Rs 5000                     confidence=0.97
  - Amazon Pay                  confidence=0.93

Example 3
--------------------------------------------------------------------------------
Input SMS:
Recharge successful for mobile number 9876543210.

Extracted Keywords:
  - Recharge successful         confidence=0.96
  - mobile number               confidence=0.94
  - 9876543210                  confidence=0.98

Example 4
--------------------------------------------------------------------------------
Input SMS:
Congratulations! You have won a free iPhone. Click the link to claim now.

Extracted Keywords:
  - Congratulations             confidence=0.95
  - free iPhone                 confidence=0.97
  - Click the link              confidence=0.94
  - claim now                   confidence=0.92

Example 5
--------------------------------------------------------------------------------
Input SMS:
Reminder: your electricity bill of Rs 1240 is due tomorrow.

Extracted Keywords:
  - electricity bill            confidence=0.96
  - Rs 1240                     confidence=0.97
  - due tomorrow                confidence=0.95

================================================================================
```